

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1461

New Salisbury 138 kV
July 04, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1461. The Interconnection Customer proposes a 155.8 MW solar photovoltaic facility interconnecting at the New Salisbury 138 kV substation in the Virginia area. The request was received on April 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 516 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,033,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dpc Prairie Du Chien / Lorin Industries 138kV	Dpc Prairie Du Chien / Tampa Bay 138kV	129.5	205	18.6
Lorin Industries / Tampa Bay Substation 138kV Line	Dpc Prairie Du Chien Substation / Tampa Bay 138kV	119.9	211	18.6
Chatom / Alvin 138kV	Sans Souci / Oak Crk Station 138kV L	116.4	208	15.6
Tampa Bay / Roslyn 138kV	Page / Purdy 138kV line	113.5	327	25.0
Dpc Prairie Du Chien / Tampa Ay 138kV	Page / Purdy 138kV Line	112.9	272	20.1
Tes Filer City Sta / Edinburg 138kV	Page Substation / Purdy Substation 138kV	111.9	203	15.8
Roslyn / Buckhorn 138kV	S Bluefield / Roslyn 138kV	109.6	140	13.1
Lewisburg / Siloam Springs 138kV	Ft Benton / Detour 138kV line	106.7	356	5.9
Springfield Power Llc Substation / Edinburg Substation 138kV	Page / Purdy 138kV	103.2	231	15.8
S Bluefield / Springfield Power Llc 138kV	Page / Purdy 138kV L	102.4	252	15.8
Frito Ay / Timberville 138kV	Chatom Substation / Purdy 138kV	102.2	274	45.2
Roslyn / South Bluefield 138kV line	Page / Purdy 138kV	102.1	260	15.8
Timberville / New Salisbury 138kV	Purdy / Chatom 138kV line	102.0	280	45.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dpc Prairie Du Chien / Lorin Industries 138kV	Rebuild line with higher-capacity conductor	\$3,532,000	26% (\$907,000)
Lorin Industries / Tampa Bay Substation 138kV Line	Rebuild line with higher-capacity conductor	\$5,304,000	26% (\$1,362,000)
Chatom / Alvin 138kV line	Rebuild line with higher-capacity conductor	\$3,889,000	12% (\$454,000)
Roslyn / Tampa Bay 138kV	Rebuild line with higher-capacity conductor	\$9,594,000	24% (\$2,283,000)
Dpc Prairie Du Chien / Tampa Bay 138kV	Rebuild line with higher-capacity conductor	\$4,161,000	27% (\$1,131,000)
Tes Filer City Sta / Edinburg 138kV	Rebuild line with higher-capacity conductor	\$14,235,000	28% (\$3,993,000)
Roslyn / Buckhorn 138kV	Reconductor line	\$2,962,000	28% (\$836,000)
Lewisburg / Siloam Springs 138kV	Reconductor line	\$2,945,000	8% (\$241,000)
Springfield Power Llc Substation / Edinburg Substation 138kV	Reconductor line	\$1,434,000	28% (\$402,000)
S Bluefield / Springfield Power Llc 138kV	Reconductor line	\$726,000	28% (\$204,000)
Frito Lay / Timberville 138kV	Reconductor line	\$2,339,000	42% (\$976,000)
Roslyn / S Bluefield 138kV	Reconductor line	\$2,742,000	28% (\$769,000)
New Salisbury / Timberville 138kV	Reconductor line	\$1,137,000	42% (\$475,000)
	Total allocated to project		\$14,033,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.